Yu Yuan

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Research Interests

Physics-Consistent and Controllable Video Generation, Physical World Models, Generative Models, Multimodal AI, Agentic AI, 3D/4D Generation, Computational Photography, Image/Video Editing/Enhancement, VLM.

Education

Purdue University — West Lafayette, IN, US
PhD Student in Electrical and Computer Engineering *Advisor: Stanley H. Chan* — *08/2023 – 01/2027 (expected)*

Shanghai Jiao Tong University — Shanghai, China
MS in Aeronautical and Astronautical Science and Technology — *09/2020 – 06/2023*

Shanghai Jiao Tong University — Shanghai, China
BEng in Aeronautics and Astronautics Engineering — *09/2016 – 06/2020*
Minor Degree in Administration Management — *10/2018 – 06/2020*

Experience

Roblox — San Mateo, CA
Research Intern (Foundation AI, Video World Models Team) — *09/2026 – Present*

- Developing real-time video foundation models, with a focus on **improving visual quality.**
- Exploring **recursive self-improvement (RSI)** for video editing.

Adobe Research — San Jose, CA
Research Intern (Multimodal Foundation Models Team) — *05/2026 – 08/2026*

- **Led the 0-to-1 development of a novel causal video dynamics editing framework RVD**, driving the full stack from problem formulation and large-scale data generation to video foundation model training, evaluation, and system integration.
- **Built a scalable multimodal data engine** that automatically generated paired counterfactual videos through data sourcing, filtering, VLM-based edit planning, video generation, quality control, and benchmark construction.
- **Delivered the first general-purpose framework for scene dynamics editing** while preserving scene consistency, resulting in 1 research paper submission and 2 patent filings.

Purdue University — West Lafayette, IN
Research Assistant (Intelligent Imaging Lab) — *08/2023 – Present*

- **Led multiple end-to-end research and engineering projects** in video generation, world modeling, physics-aware generative AI, and computational photography, spanning problem formulation, model development, large-scale training, evaluation, and open-source release.
- **OptiWorld – Optimal-Controlled Video World Generation**: Introduced optimal control into video world generation, jointly optimizing object dynamics under physical, safety, efficiency, and smoothness constraints for controllable trajectory generation.
- **SeeU – 4D Understanding and Video Generation**: Developed a unified continuous 4D world model that learns spatiotemporal dynamics from 2D observations, enabling generation and editing at unseen times and viewpoints.
- **NewtonGen – Physics-Consistent and Controllable Video Generation**: Developed a physics-consistent and controllable and controllable video generation framework with Neural Newtonian Dynamics, unifying diverse physical dynamics through neural ODEs.
- **Generative Photography – Camera-Controlled Image/Video Generation**: Introduced a camera-physics-aware generative photography paradigm for scene-consistent control over aperture, focal length, shutter speed, and color temperature.
- **Authored 5 first-author papers** at top-tier venues including **ICLR, CVPR, and NeurIPS**, spanning video generation, world modeling, physical dynamics, and controllable generative models.

Selected Publications

Yu Yuan, Yawen Lu, Guoxian Song, Kevin Duarte, Ratheesh Kalarot, Di Chang, Xijun Wang, Stanley H. Chan.
Reimagine Video Dynamics
2026 [Under Review]

Yu Yuan, Jianhao Yuan, Xijun Wang, Daiqing Li, Liu He, Lu Ling, Stanley H. Chan.
OptiWorld: Optimal Control for Video World Generation under Physical Constraints
NeurIPS 2026

Jianhao Yuan, **Yu Yuan**, Benjamin Ramtoula, Lukas Vierling, Paul Newman, Lars Kunze, Philip Torr, Daniele De Martini
Beyond Visual Quality: A Study of Test-Time Planning with World Action Models
2026 [Under Review]

Yu Yuan, Tharindu Wickremasinghe, Zeeshan Nadir, Xijun Wang, Yiheng Chi, Stanley H. Chan.
SeeU: Seeing the Unseen World via 4D Dynamics-aware Generation
CVPR 2026

Yu Yuan, Xijun Wang, Tharindu Wickremasinghe, Zeeshan Nadir, Bole Ma, Stanley H. Chan.
NewtonGen: Physics-Consistent and Controllable Text-to-Video Generation via Neural Newtonian Dynamics.
ICLR 2026

Yu Yuan, Xijun Wang, Yichen Sheng, Prateek Chennuri, Xingguang Zhang, Stanley H. Chan.
Generative Photography: Scene-Consistent Camera Control for Realistic Text-to-Image Synthesis.
CVPR 2025 [Highlight & Demo]

Yu Yuan, Yiheng Chi, Xingguang Zhang, Stanley H. Chan.
iHDR: Iterative HDR Imaging with Arbitrary Number of Exposures.
ICIP 2025

Xingguang Zhang, Nicholas Chimitt, Xijun Wang, **Yu Yuan**, Stanley H. Chan.
Learning Phase Distortion with Selective State Space Models for Video Turbulence Mitigation.
CVPR 2025 [Highlight]

Joonyeoup Kim, **Yu Yuan**, Xingguang Zhang, Xijun Wang, Stanley H. Chan.
Astrophotography Turbulence Mitigation via Generative Models.
ICIP 2025

Yu Yuan, Jiaqi Wu, Lindong Wang, Zhongliang Jing, Henry Leung, Shuyuan Zhu, Han Pan.
Learning to Kindle the Starlight.
arXiv 2022

Yu Yuan, Jiaqi Wu, Zhongliang Jing, Henry Leung, Han Pan.
Multimodal Image Fusion based on Hybrid CNN-Transformer and Non-local Cross-modal Attention.
arXiv 2022

Patents

Learning a Compact Dynamics Token for Precise Video Motion Reconstruction and Editing. [Filing].

A Data Pipeline for Generating Counterfactual Video Pairs from Real Videos. [Filing].

Trajectory resolving and registering method based on autonomous underwater robot inertial navigation and ultra-short baseline positioning sensor. CN 114993313 A.

Awards & Competitions

Ivy League Scholarship for Exceptional Students (Innovation)	Top 1%
Outstanding Graduate of Shanghai Jiao Tong University	Top 10%
Shanghai International Creators Competition: Special Competition of Drones	First Prize
DELL AI for Social Innovation Competition: AI/ADAS of Intelligent Car	National Second Place

Other Activities

Teaching Assistant, Automatic Control Theory, Shanghai Jiao Tong University

Workshop Organizer, WACV GAIP 2026

Reviewer, CVPR, ICCV, ECCV, ICLR, NeurIPS, WACV, CVPRW, ICIP, ICASSP

IEEE Graduate Student Member

Skills

Languages: English (Professional Working Proficiency), Chinese (Native)

Programming: Python, PyTorch, HTML, JavaScript, Matlab, C/C++ (Basic)

Technologies: Pre-training, SFT, Post-training, LoRA Finetuneing, Diffusers, Transformers, 3DGS, CUDA, OpenCV, NumPy, Git, LaTeX, Linux, Slurm, Markdown, Gradio, NVIDIA Jetson, Adobe Photoshop, Adobe Premiere

Others: Photography, Fishing, Drones, Rowing